

Hapti-met: A Construction Helmet with Directional Haptic Feedback for Roadway Worker Safety

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Abstract. Construction workers account for 20% of pedestrian fatalities around roadway work zones in the United States. Recent innovations in wearable warning devices roadway construction safety have proposed activating haptic (i.e., vibration motor) notifications in safety vests to alert workers of the direction of hazardous vehicles (speeding, collision). However, the vests' embedded motors may lose effectiveness when worn outside of heavy clothing (i.e., winter clothes). This study introduces a novel safety helmet wearable, "Hapti-met," that similarly employs direction-based notifications with vibration motors activated directly onto a roadway worker's head. The study also proposes a method for evaluating the wearable's impact on workers' life safety, utilising a virtual reality (VR) environment where human roadway workers must work while avoiding simulated collision vehicles. Preliminary results show a 9.8% reduction in collisions with vehicles when workers receive haptic notifications from Hapti-met compared to when they do not.

1. Introduction

Construction workers in the United States (US) account for 20% of pedestrian fatalities around roadway work zones (ARTBA 2023). To ensure worker safety on roadways, federal and state level transportation agencies provide safety training guidelines and standards primarily advising on how to set up work zones with signage and traffic control devices (i.e., cones, barricades) prior to construction activities (Federal Highway Administration 2022). Safety standards also stipulate that workers must wear bright-colour personal protection equipment (PPEs), such as hard-hat helmets and safety vests. These safety measures are intended to improve the attention of vehicle motorists as they approach a work zone. With respect to what workers should do during construction, these safety training guidelines merely advise workers stay vigilant and attentive of nearby traffic around themselves and other workers. To assist workers in their attention to traffic, numerous work zone intrusion alert (WZIA) technologies have emerged in recent years, designed to remotely sense hazardous vehicles (e.g., speeding, collision) and promptly alert workers with sounds and flashing lights from a stationary alarm source (e.g., loudspeaker, lamps) (Awolusi and Marks 2019). A notable limitation in these technologies is the fact that workers tend either not notice or respond slowly to these alarms as they stand farther from WZIA alarm sources (Gambatese et al. 2017; Mishra et al. 2021). With recent advances in Internet of Things (IoT), intelligent computing applications such as wearable warning devices have been proposed to counteract this limitation by integrating sensors and alarm sources into construction workers' PPEs (Yadav et al. 2022). Many proposed wearable warning devices (e.g., smart glasses, remote speakers) rely on conveying visual and auditory cues to alert workers but can fail in construction sites with naturally high ambient levels of noise and visual complexity (Aliyev et al. 2020; Baek and Choi 2020; Golovina et al. 2021).

As an alternative, haptic-based (i.e., vibrotactile motors) wearable warnings are more likely to be sensed by roadway workers compared to alarms of other modalities (e.g., audio) (Qin et al. 2022). Recent roadway worker-centred studies have proposed embedding vibration motors into

safety vests (Sakhakarmi et al. 2021; Sakhakarmi and Park 2022; Yang and Roofigari-Esfahan 2023), where the location of the vibration can represent different meanings (e.g., direction of an approaching vehicle, direction a worker should move). On the other hand, wearable studies investigating safety vests and other haptic applications (e.g., driver seats) have noted vibration motors' limited effectiveness when people are wearing thick winter clothing layers between the motors and their skin (Duthoit et al. 2016; Golovina et al. 2021; Yang and Roofigari-Esfahan 2023). There is a need to investigate how to integrate haptics into other PPE that are in direct contact with a worker's body in all seasons and climates. With a wide range of applications explored in other industrial and transportation safety challenges (i.e., factory worker, cyclist), helmets can be a better choice of PPE to integrate with haptic warning devices for roadway workers (Vo et al. 2021; Chen et al. 2023). Furthermore, past studies developing haptic warning devices tend to only measure how quickly workers notice the vibrations (i.e., clicker reaction time) and other aspects of workers' subjective experience (i.e., user preference, usability rating), leaving unaddressed whether those devices can actually reduce the risks of worker injury and fatality from work zone intrusions (Gambatese et al. 2017; Yang and Roofigari-Esfahan 2023). Field testing the impact of warning devices on workers' life safety risks in real world roadway work zones is a challenging endeavour, requiring careful consideration of the safety of all participants involved. With the recent ubiquity of immersive virtual reality (VR) systems, there is an opportunity to realistically and safely simulate these roadway safety risks and evaluate whether wearable warning devices can actually improve worker's attention and reduce their life safety risks.

This study introduces Hapti-met, a novel system of vibration motor modules designed to be attached to different parts of a safety helmet's (i.e., hard hat) interior headband and anti-concussion harness for alerting roadway construction workers of hazardous vehicles. The wearable device can either employ a regular notification strategy where a single vibration motor activates at the top centre (i.e., apex) of a worker's head or a directional notification strategy where one of four vibration motors on different sides of a worker's head will vibrate to indicate the direction (i.e., front, back, left, right) of an approaching vehicle relative to where the worker's head is facing (i.e., head orientation).

The main contributions of this study are: 1) an outline of Hapti-met's design details, 2) a preliminary implementation of Hapti-met's regular and directional haptic notification strategies towards reducing the risks of collisions with vehicles intruding roadway work zones. This study leverages a previously developed integrated traffic simulation and virtual reality (VR) platform where human workers can perform simulated roadway construction activities while hazardous vehicles occasionally intrude the work zone from different directions (Ergan et al. 2022). Wearable user study trials on the VR platform test the hypothesis that workers can avoid more vehicle collisions when the Hapti-met employs a directional notification strategy than when it employs a regular notification strategy or does not activate any notifications at all (i.e., baseline). By contributing a novel wearable warning device and innovative approach for evaluating the wearable's impact on worker life safety risks, this study hopes to guide the design of future wearable warning devices that ultimately prevent roadway work zone accidents.

2. Related Works

2.1 Haptic Wearable Devices for Roadway Worker Safety

The use of haptic feedback mechanisms (e.g., vibrotactile motors) for human workers has been explored in past construction safety studies (Simeonov et al. 2011; Yadav 2017; Aliyev et al.

2020). Most of them assume a generic vertical construction site (i.e., buildings), leaving unconsidered important risk factors more pertinent to horizontal construction sites (i.e., roadways, highways), such as speeding traffic vehicles. With respect to roadway construction workers, past studies evaluated the effects of integrating different vibration motor configurations (e.g., motor spacing, vibration patterns) into safety vests to alert highway construction workers if they left the work zone area or a vehicle has intruded the work zone (Roofigari-Esfahan et al. 2021; Sakhakarmi and Park 2022; Yang and Roofigari-Esfahan 2023). Yang and Roofigari-Esfahan particularly investigated how well highway workers interpreted vibrations following a directional notification strategy (i.e., vibrations indicate direction of hazard), but mainly measured the number of correct interpretations and aspects of workers' subjective experience (e.g., usability ratings). As an alternative to safety vests, this paper will investigate the potential benefits of implementing a directional vibration strategy on construction helmets. Since government agencies and PPE manufacturers prohibit any clothing underneath the helmet that limit its "protective properties," hard hats and integrated haptic vibration motors are often likely in direct contact with workers' heads (OSHA 2006). Furthermore, recent studies in both the construction and transportation safety domains have demonstrated the wide-ranging potential of head-mounted wearable warning devices, including hard hats and cyclist helmets (Aliyev et al. 2020; Campero-Jurado et al. 2020; Vo et al. 2021; Chen et al. 2023; Ma et al. 2024). This growing body of work motivates this study to explore other ways haptics in safety helmets can augment the senses of construction workers to be more vigilant of traffic vehicles. Given the lack of research on urban roadways compared to highways, this study will also focus on short term roadwork scenarios in its evaluation of the haptic warning device (Ozbay et al. 2012; Ergan et al. 2022).

2.2 VR Studies on Roadway Worker Safety

The increasing commercial availability of virtual reality (VR) technologies has led to a fast-growing field of research developing immersive applications for construction safety. Past studies tend to focus on VR for human worker safety training and developing automation solutions for monitoring their physiology (i.e., biometrics) and behaviour (Moore and Gheisari 2019; Kim et al. 2023). Towards the specific domain of transportation and roadway (i.e., street, highway) construction safety, recent studies have used the sensors in the VR headset to track workers' head orientation and collisions with nearby construction equipment (e.g., asphalt paver) (Kim et al. 2021). This study builds on those previous studies by investigating how roadway workers will move in reaction to an approaching traffic vehicle, including whether they can successfully avoid collisions.

Along with the ubiquity of Internet of Thing-enabled (IoT) technologies, recent studies are utilising VR platforms to investigate how worker safety may improve with "hardware in the loop" including wearable devices, rather than just focusing on the human workers themselves (Simeonov et al. 2011; Ergan et al. 2022; Qin et al. 2022; Esfahani et al. 2024). A prior study by the research team investigated the impact of a smartwatch raising different sound and haptic vibration alarm patterns on how workers physically reacted and saw a vehicle in their field of view (Qin et al. 2022). Data collection in that study was enabled with custom scripting in the VR game engine that could track vehicle locations and worker's head orientation (i.e., gaze direction). Other recent studies in the transportation domain have used VR to develop wearable warning devices for roadway workers (Chan et al. 2020; Yang and Roofigari-Esfahan 2023). Yang and Roofigari-Esfahan particularly employed a 360° video recording of a highway construction site as the immersive VR experience for testing a safety vest's haptic notifications, arguing the recording is more photorealistic than visualization in other VR game engine-based studies. The haptic notifications and workers' movements, however, were manually triggered

and recorded by research staff. While photorealism is a noteworthy advantage of 360° video recordings, this study will employ a VR game engine for more precise control over how hazardous vehicles approach work zones and automate the data collection on how workers will react to those vehicles (e.g., vehicle collision rates). Using this data collected from a VR game engine platform, this study estimates how a haptic wearable warning device can assist in preventing worker collisions with traffic vehicles, thereby mitigating their life safety risks.

3. Research Method

3.1 Hapti-met Prototype Design

Hapti-met features four coin vibration motors (2-5V) integrated into a safety helmet's interior headband and anti-concussion harness, one for each cardinal direction (front, back, left, right) on a worker's head (Figure 1a). A fifth vibration motor is positioned at the centre of the worker's head (i.e., apex) to provide haptic-based alerts with no directional associations. Two types of 3D printed clip modules are designed to readily attach and detach the coin vibration motors from the helmets. The first module type is designed to attach to the front and back of the helmet via Velcro fasteners. The second type can clip onto the helmet harness straps to be positioned on the head's centre (i.e., apex), left, and right sides. In the event of motor malfunction or damage, users can effortlessly replace the motor modules and reattach them to the helmet. The Hapti-met's intended operation can follow a regular notification strategy where only the centre motor vibrates when a hazardous vehicle approaches the roadway work zone. Alternatively, it can follow a directional notification strategy where one of four cardinal direction's vibration motors can alert the worker of an approaching hazardous vehicle's direction relative to where that individual worker's head is currently facing (i.e., gaze direction). The Hapti-met's intended design is to have these five motors powered by a small control board with WiFi capabilities (e.g., ESP32) and lightweight battery (e.g., 3.7V Li-Po) attached to the back of the helmet, similar to previous helmet-based studies (Aliyev et al. 2020).

The initial prototype developed in this study consists of significant differences from the intended design. The prototype uses a much larger control board (Raspberry Pi A+) and battery pack (10Ah Power Bank) (Figure 1b). For this initial prototype, a single alert involves vibrating the motor for a duration of 350ms, repeated twice (2x), with a 250ms pause in between, since this vibration pattern was found to be noticeable by human workers in prior studies by the research team (Qin et al. 2022). For VR user studies conducted in this study, the headband and

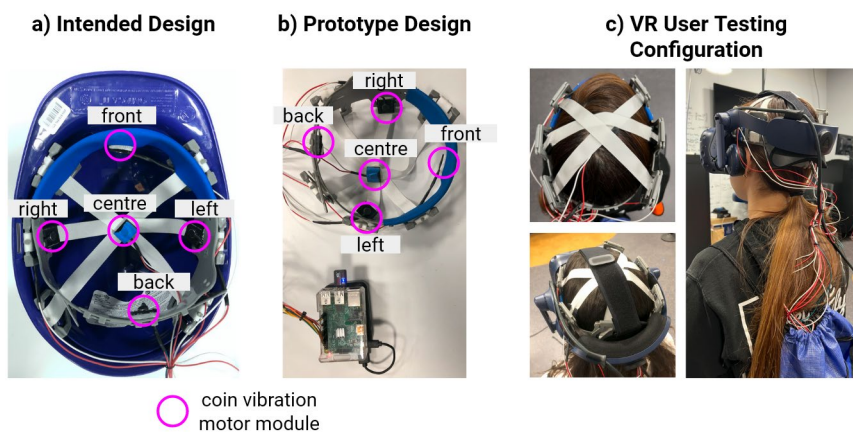


Figure 1: Hapti-met's intended design, prototype design, and configuration with VR headset.

harness are removed from the helmet so that the prototype device can be comfortably worn along with a VR headset (Figure 1c). As will be discussed in subsequent sections, this study relies on the VR headset and VR game environment to track where a hazardous vehicle is located relative to a worker's real-time head orientation. In practice, the intended helmet design would need to include additional sensors (e.g., gyroscope) for tracking a worker's head orientation while other sensor systems (e.g., camera, ultrasonic) would need to be installed in work zone site layouts to detect hazardous vehicle locations. While additional design iterations are expected before Hapti-met is ready for field deployment (e.g., safety standards, component durability), the prototype version utilized in this study is sufficient for the purposes of evaluating its directional notification capabilities.

3.2 VR Testbed for Evaluation of Haptic Notification Strategies

VR Roadway Work Zone Collision Testbed

To evaluate the effects of Hapti-met's possible notification strategies on rates of worker-vehicle collisions, this study leverages a previously developed integrated VR and micro-traffic simulation platform. In this platform, the Unity VR environment consists of work zone in an urban street where workers are setting up a short-term work zone (i.e., lane closure) and vehicle trajectories are simulated using real-world traffic data in a micro-traffic simulator (i.e., SUMO simulation of urban mobility tool) (Ergan et al. 2022). Previous work utilized this platform to investigate how workers react to randomized sound and haptic notifications on a smartwatch triggered by dangerous vehicles (e.g., intrusion, speeding) (Qin et al. 2022). While notifications in previous studies did not change depending on any properties of the dangerous vehicles (e.g., direction, speed), this study modifies the VR platform's simulations to examine worker reactions to a helmet's haptic notifications that change depending on the direction of the dangerous vehicle's incoming trajectory.

This study tests the Hapti-met in a roadway work zone scenario where an individual construction worker is tasked with placing cones on an urban street for a short-term work zone. User study participants can use the VR controller to pick up the cone objects and place them at locations indicated by spherical white lights in the VR simulation (Figure 2a). As participants complete this task, SUMO-controlled normal traffic vehicles will slowly pass by the work zone area. At the same time, the Unity VR game engine can generate fast collision vehicles to randomly appear and intrude the work zone from the north or south side (Figure 2b). Normal vehicles appear around once every 60 seconds and move at 11 m/s (25 mph). Collision vehicles will appear once every 14 seconds on either side, when normal vehicles are not present near the work zone and move at 29 m/s (65 mph). Both the vehicles and the VR user have virtual 3D colliders attached, a Unity game engine component that detects whether 3D objects intersect one another. In effect, the VR platform can record whether the roadway worker successfully

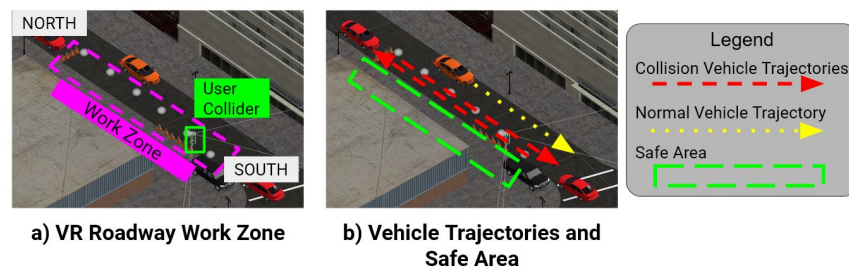


Figure 2: Configuration of VR environment for roadway work zone and vehicle trajectories.

avoided a collision with an intruding vehicle. While participants can use the VR controller to shift their user 3D collider in parallel with the road axis and complete the cone placements, they cannot do the same with the VR controller to move perpendicular to the road. Thus, participants must move with their physical bodies to move into a safe area and avoid collisions with vehicles. As a testbed for evaluating Hapti-met, the VR platform also has a mechanism for triggering alarms on wearable warning devices. When they appear in the VR environment, the collision vehicles immediately send an alarm signal to the Hapti-met prototype to activate a vibration motor (Figure 3). Depending on Hapti-met's notification strategy setting, one of the five vibration motors will activate. For the regular notification strategy, only the centre motor will be activated. For the directional notification strategy, one of four motors (front, back, left, right) will be activated depending on the user's head orientation tracked by the VR headset sensors and the location of the vehicle detected by the VR game engine when the alarm is triggered. Along with reaction data investigated in previous publications, this study will specifically collect data on whether the worker collides with the intruding vehicle or not (Qin et al. 2022). If they fail to avoid collisions, workers temporarily see the word "Collision" in red text in their VR headset display but can continue working to place the cones (Figure 3, lower right). Overall, the VR platform in this study can realistically simulate an urban roadway work zone and quantitatively measure the impact of Haptimet's different notification strategies in helping workers avoid collisions with hazardous vehicles.

Initial Data Collection from Wearable User Studies in VR

As a demonstration of the roadway worker collision testbed, this study conducted an initial set of human subject user studies with Hapti-met and the VR platform, as approved by New York University IRB (IRB-FY2020-3946). Twelve ($n=12$) affiliated NYU students participated in the user studies, ages between 23 and 36, with 7 identifying as male (58.3%) and 5 identifying as female (41.7%). While all participants were university students, a considerable percentage ($n = 46.2\%$) of them reported prior experience in general construction and industrial site experience, ranging between 6 months to 5 years. Though user studies with more experienced roadway workers are planned for future work, findings from this initial participant cohort can start to reveal how humans generally react to different haptic notification strategies. Each participant's user study session began with putting on the Hapti-met wearable prototype design and HTC Vive Pro VR headset (see Figure 1c). Prior to data collection trials in the VR simulation, each participant was given a tutorial on how to use the VR interface, including how to use controllers to complete the task of placing cones. While participants were instructed on their restrictions for moving with the VR controller and allowed to move with their physical bodies, research staff did not give any specific advice on how to avoid the collisions in the tutorial. Once participants felt comfortable with the VR interface, three types of data collection

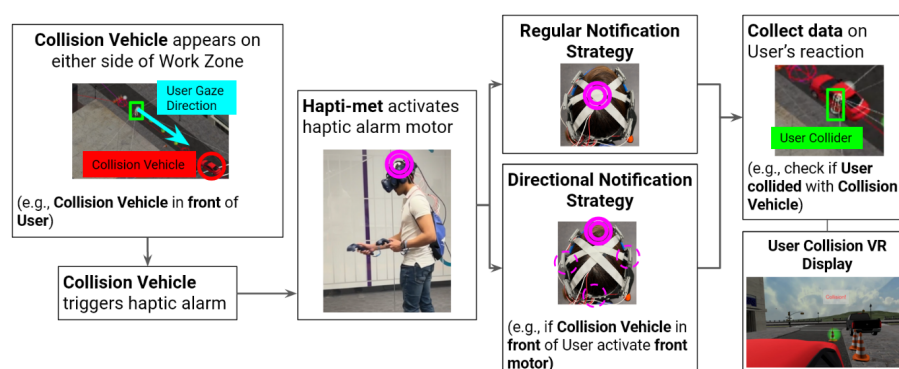


Figure 3: VR platform for collecting data on user's reactions to collision vehicles and haptic alarms.

trials were conducted: one with no haptic alarms for collision vehicles, one with Hapti-met using a regular notification strategy, and one using a directional notification strategy. Before performing each notification strategy trial, participants received a brief demonstration of that notification strategy. The order in which the three trial conditions (no alarms, regular notification strategy, directional notification strategy) were randomized for each participant to mitigate potential order effects in measurements of worker-vehicle collision rates. Each trial concluded once the participant successfully completed placing cones on all spherical white lights. After the three trials, participants filled out a survey regarding their experience with both the VR interface and Hapti-met wearable device's notification strategies. During data collection trials, various types of data (e.g., head position and orientation coordinates) were collected to capture participant reactions to the alarms, including those explored in previous publications (Qin et al. 2022). In this paper, the primary data of interest is whether participants collided with each hazardous vehicle under the different trial conditions (no alarms, regular notification, directional notification). Specifically, the number of collision vehicles participants experienced as they placed cones and whether they collided with each vehicle was recorded for each trial. Results present preliminary data analysis on how notification strategy may have impacted the rate at which participants avoided collisions with vehicles during this initial stage of data collection.

4. Initial Results

A total of 314 collision vehicles were tested across the participant's trials, with 79 collisions observed overall. Since the length of time of each participants' trials depended on when they completed the tasks, the total number of collision vehicles tested for each trial condition (no alarms, regular notification strategy, directional notification strategy) varied: 114 collision vehicles tested with no alarms, 99 collision vehicles tested with centre vibrations (regular strategy), and 101 collision vehicles tested with directional vibrations. Additionally, of the total 130 and 184 vehicles that randomly appeared north and south of the work zone, 102 (78.4%) north vehicle collisions and 133 (72.3%) south vehicle collisions were avoided. This suggests that participants had a slight disadvantage in avoiding vehicles that moved against the direction of normal vehicle traffic. The rate at which participants avoided collisions with vehicles differed with each type of trial condition (Figure 4). Further analysis into the distribution of collision avoidance rates for each individual data collection trial is shown in Figure 5 box plots. Participants demonstrated a 68.4% avoidance rate when no alarms were provided. In comparison, the regular and directional vibration strategies yielded a higher avoidance rate of

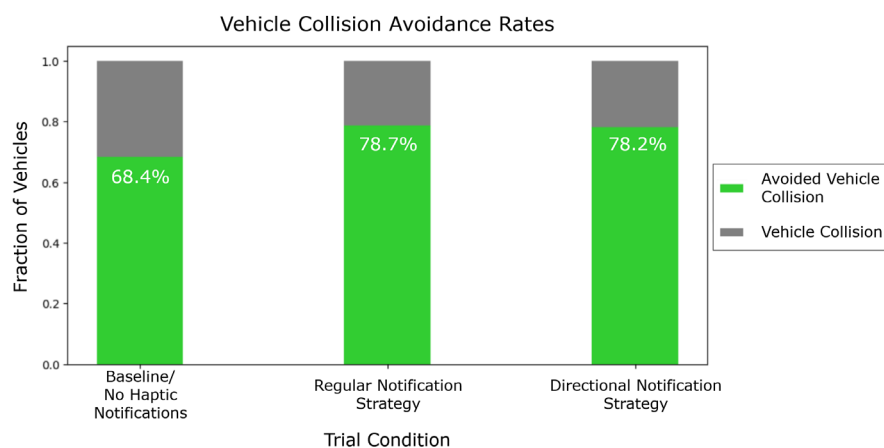


Figure 4: Rates at which participants avoided collision with vehicles under different trial conditions.

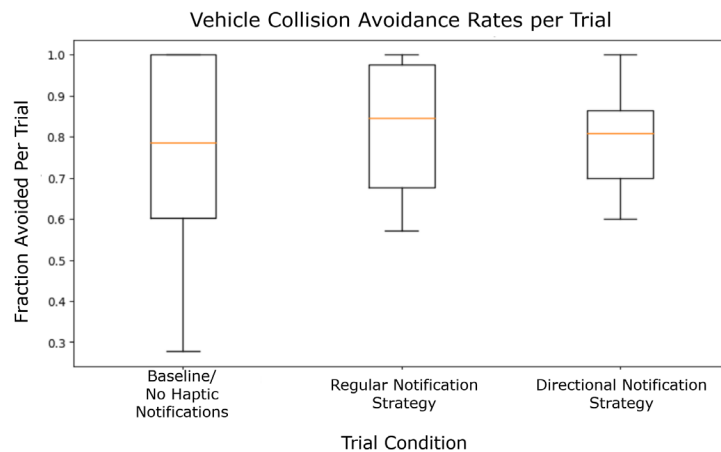


Figure 5: Rates at which participants avoided collision with vehicles within each trial.

78.7% and 78.2%, respectively (Figure 4). In Figure 5, for the baseline condition with no haptic alarms, the median avoidance rate is 78.6%, with a range spanning from 27.7% to 100%, and an interquartile range (IRQ) of 39.8%. In comparison, the regular notification strategy shows a median avoidance rate of 84.5%, with a narrower range from 57.1% to 100% and a smaller IRQ of 29.9%. Similarly, the directional vibration strategy exhibits a median avoidance rate of 80.9%, with a range from 60% to 100% and a IRQ of 16.5%. In particular, Figure 5 displays smaller ranges and IRQ of both regular and directional notification strategies compared to the baseline trial condition (i.e., no haptic alarms), conveying that workers can avoid collisions more consistently when vibrations alert them of an approaching hazardous vehicle. These findings reinforce the effectiveness of both notification strategies in enhancing collision avoidance performance among participants, but it is unclear with this limited dataset whether a directional or regular notification strategy is more effective.

5. Discussion and Future Work

Preliminary results indicate that both regular and directional notification strategies can help roadway workers avoid collisions with hazardous vehicles that intrude the work zone. Findings remain inconclusive if the directional notification strategy provided significant benefit compared to the regular notification strategy. Though Figure 4 shows that participants could avoid more than half of all collision vehicles in all trial conditions, other data suggests participants had a more difficult time avoiding collisions when a vehicle approached the work zone in a direction opposite to normal traffic flow. Based on qualitative survey comments from user study participants, several possible explanations for the observed trend are provided. With regards to the collision rates observed when no haptic alarms were presented, some participants noted that the sound of the approaching hazardous vehicle provided adequate spatial audio cues for where the vehicle was coming from and when to start moving to a safer area. That said, survey responses indicate that participants perceived (i.e., self-report) they felt a vibration ahead of hearing the sound of the vehicle. With respect to the directional notification strategy, participants also reported that this way of conveying directional information introduced confusion and caused delays in their reactions. One participant remarked “it takes a couple of milliseconds to process the direction and then make an action,” suggesting that directional vibrations on different sides of a worker’s head require additional mental load to interpret. Conversely, other participants expressed preference for a single vibration stating that they “focused on being safe, rather than the direction” of the approaching vehicle, indicating they moved without trying to interpret directional vibrations. Additionally, certain participants with

longer hair reported difficulty in noticing vibrations in the back side of their head. Further investigations into how Hapti-met can work with different hair styles (e.g., afros) and head garments (e.g., balaclavas) are warranted. Nonetheless, the initial results suggest benefits of safety helmets employing haptic notifications and should encourage further investigation into the potential utility of head-mounted wearables that provide haptic feedback.

6. Conclusion

This paper presents a preliminary development and evaluation of a Hapti-met warning device, which provides vibrations on different sides of a roadway worker's head to alert them of the direction of an approaching hazardous vehicle. In VR-based user studies that compared these collision rates between trials without any alarms and trials where participants received haptic vibration notifications, initial findings show that Hapti-met's vibrations helped workers to avoid collisions more consistently. Compared to trials where participants only felt a vibration at the top-centre (i.e., apex) of their head, vibrations that indicate the direction of a vehicle may incur greater mental load for a worker to interpret and delay their reaction. Given this paper's limited sample of participants, future work will conduct user studies encompassing a more inclusive range of user demographics and attire (i.e., different hair styles, head garments, roadway construction experience). Overall, this study contributes a novel integration of haptic warning devices with safety helmets with experiment findings that motivate further research in its applications for roadway worker safety.

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